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BY MALCOLM JAMESON



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WILL E. R. RAY, EDITOR

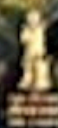
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*** START OF THE PROJECT GUTENBERG EBOOK BEAM PIRATE

Beam Pirate

By GEORGE O. SMITH

Illustrated by Alfred

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Mark Kingman was in a fine state of nerves. He looked upon life and the people in it as one views the dark-brown taste of a hangover. It seemed to him at the present time that the Lord had forsaken him, for the entire and complete success of the solar beam had been left only to Venus Equilateral by a sheer fluke of nature. Certainly he, nor anyone else, could have foreseen the Channing Layer, that effectively blocked any attempt to pierce it with the strange, sub-level energy spectrum over which the driver tube and the power-transmission tube worked, representing the extremes of the so-called spectrum.

But Venus Equilateral, for their part, were well set. Ships plied the spaceways using their self-contained power only during atmospheric passage, and paid Venus Equilateral well for the privilege. The Relay Station itself was powered on the solar beam, and the costly shipments of potential power had been stopped. There were other relay stations that belonged to the communications company; Luna, Deimos and Phobos, and the six that circled Venus in lieu of a satellite; all were powered by the solar beam. And the solar observatory on Mercury used but little power, so the needs of the observatory became the sole income for Terran Electric's planetary rights of the solar beam, since Mercury owned no air of its own.

Mark Kingman was beginning to feel the brunt of Channing's statement to the effect that legal-minded men were of little importance when it came to the technical life in space, where men's lives and livelihood depended more on technical skill than upon the legal pattern set for their protection in the complex society of planetary civilization.

It seemed that way. For instead of gaining their ends by legal restrictions on the power-transmission tube investigations, Terran Electric had lost their chance. Venus Equilateral had the legal right to tinker with the transmission tubes all they wanted to, and in return, Terran Electric held all of the planetary rights to Venus Equilateral's solar beam—which in the domain covered by natural celestial bodies was about as valuable as the gold-mining rights to the crater Tycho.

And everyone knows that Luna, as a valuable piece of real estate, is useful only to Venus Equilateral as a place to plant the Lunar Relay Station that handled the Terran Beam and punched downward at the Heaviside Layer.

Luna's valuable assets as to mineral rights consisted of a bit of talc—no longer used because of plastic engineering—and pumice—no longer used because of synthetic engineering.

And Kingman knew that only if Terra were not abundant in granite would the Lunar granite come in handy as a source of tombstones; and that made him writhe because when he thought of tombstones he also thought of his position with Terran Electric, which had been endangered because of his own legal connivances.

He swore vengeance.

So, like the man who doggedly makes the same mistake twice in a row, Kingman was going to move Heaven, Hell, and the three planets in an effort to take a swing at the same jaw that had caught his fist between its teeth before.

Out through the window of his office, he saw men toiling with the big tube of the far roof; the self-same tube that had carried the terrific load of Venus Equilateral for ten days without interruption and with no apparent overload. Here on Terra, its output meter, operating through a dummy load, showed not the slightest inclination to leave the bottom peg and seek a home among the higher brackets.

The Channing Layer barred the passage of radiation of this so-called sub-etheric energy as effectively as the Heaviside Layer had blocked Interplanetary Communications for many, many years.

So Kingman cursed and hated himself for having backed himself into trouble. But Kingman was not a complete fool. He was a brilliant attorney, and his record had placed him in the position of Assistant Chief Attorney for Terran Electric, which was a place of no mean importance. He had been licked on the other fellow's ground, with the other fellow's tools.

He picked up papers that carried, side by side, the relative assets of Venus Equilateral and Terran Electric. He studied them and thought deeply.

To his scrutiny, the figures, seemed about equal, though perhaps the Interplanetary Communications Co. was a bit ahead.

But—he had been licked on the other fellow's ground with the other fellow's tools; he thought that if he fought on his own ground, with his own tools, he might be able to swing the deal.

And Terran Electric was not without a modicum of experience in the tools of the other fellow. His engineering department was brilliant and efficient, too; at least the equal of Channing and Franks and their gang of laughing gadgeteers. That not only gave him the edge of having his own tools and his own ground, but a bit of the other fellow's instruments too. Certainly his engineering department should be able to think of something good.

William Cartright, business manager for Venus Equilateral, interrupted Don and Walt in a discussion. He carried a page of stock market quotations and a few hundred feet of ticker tape.

Channing put down his pencil and leaned back in his chair. Walt did likewise, and said: "What's brewing?"

"Something I do not like."

"So?"

"The stock has been cutting didoes. We've been up and down so much it looks like a scenic railway."

"How do we come out?"

"Even, mostly; but from my experience, I would say that some bird is playing hooky with Venus Equilateral, Preferred. The common is even worse."

"Look bad?"

"Not too good. It is more than possible that some guy with money and the desire might be able to hook a large slice of V.E. Preferred. I don't think they could get control, but they could garner a plurality from stock outstanding on the planets. Most of the preferred stuff is in the possession of the folks out here, you know, but aside from yourself, Walt, and a couple of dozen of the executive personnel, the stock is spread pretty thin. The common stock has a lot of itself running around loose Outside. Look!"

Cartright began to run off the many yards of ticker tape. "Here, some guy dumped a boatload at Canalopsis, and some other guy glommed on to a large hunk at New York. The Northern Landing Exchange showed a bit of

irregularity during the couple of hours of tinkering, and the irregularity was increased because some bright guy took advantage of it and sold short." He reeled off a few yards and then said: "Next, we have the opposite tale. Stuff was dumped at Northern Landing, and there was a wild flurry of bulling at Canalopsis. The Terran Exchange was just flopping up and down in a general upheaval, with the boys selling at the top and buying at the bottom. That makes money, you know, and if you can make the market tick your way—I mean control enough stuff—your purchases at the bottom send the market up a few points and then you dump it, and it drops again. It wouldn't take more than a point or two to make a guy rich, if you had enough stock and could continue to make the market vacillate."

"That's so," agreed Don. "Look, Bill, why don't we set one of our Terran agents to tinkering too? Get one of our best men to try to outguess the market. As long as it is being done systematically, he should be able to follow the other guy's thinking. That's the best we can do unless we go gestapo and start listening in on all the stuff that goes through the Station here."

"Would that help?"

"Yeah, but we'd all land in the hoosegow for breaking the secrecy legislation. You know. 'No one shall ... intercept ... transmit ... eavesdrop upon ... any message not intended for the listener, and ... shall not ... be party to the use of any information gained ... et cetera.' That's us. The trouble is this lag between the worlds. They can prearrange their bulling and bearing ahead of time and play smart. With a little luck, they can get the three markets working just so—going up at Northern Landing; down at Terra; and up again at Canalopsis, just like waves in a rope. By playing fast and loose on paper, they can really run things hell, west, and crooked. Illegal, probably, since they each will no doubt claim to have all the stock in their possession, and yet will be able to sell and buy the same stock at the same time in three places."

"Sounds slightly precarious to me," objected Cartright.

"Not at all, if you figure things just right. At a given instant, Pete may be buying at sixty-five on Venus; Joe may be selling like furious at seventy-one on Mars; and Jimmy may be bucking him up again by buying at sixty-five on Terra. Then the picture and the tickers catch up with one another, and Joe

will start buying again at sixty-five, whilst Pete and Jimmy are selling at seventy-one. Once they get their periodicity running, they're able to tinker the market for quite a time. That's where your man comes in, Bill. Have him study the market and step in at the right time and grab us all a few cheap ones. Get me?"

"Sure," said Cartright. "I get it. In that way, we'll tend to stabilize the market, as well as getting the other guy's shares."

"Right. I'll leave it up to you. Handle this thing for the best interests of all of us."

Cartright smiled once again, and left with a thoughtful expression on his face. Channing picked up the miniature of the power-transmission tube and studied it as though the interruption had not occurred. "We'll have to use about four of these per stage," he said. "We'll have to use an input-terminal tube to accept the stuff from the previous stage, drop it across the low-resistance load, resistance couple the stage to another output terminal tube where we can make use of the coupling circuits without feedback. From there into the next tube, with the high resistance load, and out of the power-putter-outer tube across the desk to the next four-bottle stage."

"That's getting complicated," said Walt. "Four tubes per stage of amplification."

"Sure. As the arts and sciences get more advanced, things tend to get more complicated."

"That's essentially correct," agreed Walt with a smile. "But you're foreguessing. We haven't even got a detector that will detect driver radiation."

"I know, and perhaps this thing will not work. But after all, we've got the tubes and we might as well think them out just in case. We'll detect driver radiation soon enough, and then we might as well have a few odd thoughts on how to amplify it for public use. Nothing would tickle me more than to increase those three circles on our letterhead to four. 'Planet to Planet, and Ship to Ship' is our hope. This one-way business is not to my liking. How

much easier it would have been if I'd been able to squirt a call in to the Station when I was floating out there beyond Jove in that wrecked ship. That gave me to think, Walt. Driver-radiation detection is the answer."

"How so?"

"We'll use the detector to direct our radio beam, and the ship can have a similar gadget coupled to their beam, detecting a pair of drivers set at one hundred and eighty degrees from one another so the thrust won't upset the Station's celestial alignment. We can point one of them at the ship's course, even, making it easier for them."

"Speaking of direction," said Walt thoughtfully, "have you figured out why the solar beam is always pointing behind Sol?"

"I haven't given that much thought. I've always thought that it was due to the alignment plates not being in linear perfection so that the power beam bends. They can make the thing turn a perfect right angle, you know."

"Well, I've been toying with the resurrected heap you dropped into Lake Michigan a couple of months ago, and I've got a good one for you. You know how the beam seems to lock into place when we've got it turned to Sol, not enough to make it certain, but more than detectably directive?"

"Yep. We could toss out the motor control that keeps her face turned to the sun."

"That's what I was hoping to gain—" started Walt, but he stopped as the door opened and Arden entered, followed by a man and woman.

"Hello," said Walt in a tone of admiration.

"This is Jim Baler and his sister Christine," said Arden. "Baler, the guy with the worried look on his face is my legally wedded spouse—souse—no, spouse. And the guy with the boudoir gorilla gleam in his vulpine eye is that old vulture, Walt Franks."

Walt took the introduction in his stride and offered Christine his chair. Arden stuck her tongue out at him, but Walt shrugged it off, ignoring her. Channing shook hands with Jim Baler and then sought the 'S' drawer of his file cabinet. He found the Scotch and the soda, and then grinned: "Should have the ice under 'I,' but it's sort of perishable, and so we keep it in the refrigerator. Arden, breach the 'G' drawer and haul out the glasses, will you please? I

suppose we could refrigerate the whole cabinet, but it wouldn't sound right if people heard that we kept their mail on ice. Well—"

"Here's how, if we don't already know," said Walt, clinking glasses with Christine.

"Walt earned that 'wolf' title honestly," laughed Arden, "he likes to think. Frankly, he's a sheep in wolf's clothing!"

"What are his other attributes?" asked Christine.

"He invents. He scribbles a bit. He cuts doodles on tablecloths, and he manages to get in the way all the time," said Don. "We keep him around the place for his entertainment value."

"Why—"

"Quiet, Walter, or I shall explain the sordid details of the Walter Franks Electron Gun."

"What was that one?" asked Christine.

"You really wouldn't want to know," Walt told her.

"Oh, but I would."

"Yeah," growled Franks, "you would."

"Would you rather hear it from him, or me?" Arden asked.

"He'll tell me," said Christine. Her tone was positive and assured.

"And that'll take care of that," said Arden. "But I think we interrupted something. What were you saying about gaining, Walt?"

"Oh, I was saying that I was tinkering around the skyways with the *Anopheles*—that's the ship we hooked up with the solar beam for power, you know—and I got to wondering about that discrepancy. The faster you go, the greater is the angular displacement, and then with some measurements, I came up with a bugger factor—"

"Woah, goodness," laughed Christine. "What is a bugger factor?"

"You'll learn," said Arden, "that the boys out here have a language all their own. I've heard them use that one before. The bugger factor is a sort of multiplying, or dividing, or additive, or subtractive quantity. You perform the mathematical operation with the bugger factor, and your original wrong answer turns into the right answer."

"Is it accepted?"

"Oh sure," answered Arden. "People don't realize it, but that string of 4's in the derivation of Bode's Law is a bugger factor."

"You," said Christine to Walt, "will also tell me what Bode's Law is—but later."

"O.K.," grinned Walt. "At any rate, I came up with a bugger factor that gave me to think. The darned solar beam points to where Sol actually is!"

"Whoosh!" exclaimed Channing. "You don't suppose we're tinkering with the medium that propagates the law of gravity?"

"I don't know. I wouldn't know. Has anyone ever tried to measure the velocity of propagation of the attraction of gravity?"

"No, and no one will until we find some way of modulating it."

Jim Baler smiled. "No wonder Barney was a little wacky when he got home. I come out here to take a look around and maybe give a lift to your gang on the transmission tube—and bump right into a discussion on the possibility of modulating the law of gravity!"

"Not the law, Jim, just the force."

"Now he gets technical about it. You started out a couple of months ago to detect driver radiation, and ended up by inventing a beam that draws power out of the sun. Think you'll ever find the driver radiation?"

"Probably."

"Yeah," drawled Arden. "And I'll bet a hat that when they do, they won't have any use for it. I've seen 'em work before."

"Incidentally," asked Christine, "you mentioned the *Anopheles*, and I think that is the first ship I've ever heard of that hasn't a feminine name. How come?"

"The mosquito that does the damage is the female," grinned Jim. "The Mojave spaceyards owns a sort of tender craft. It has a couple of big cranes on the top and a whole assortment of girders near the bottom. It looks like, and is also called: *The Praying Mantis*. Those are also female; at least the ones that aren't afraid of their shadow are."

Channing said suddenly: "Walt, have you tried the propagation-time of the solar beam on the *Anopheles*?"

"No. How would we go about doing that?"

"By leaving the controls set for 1-G, and then starting the ship by swapping the tube energizing voltages from test power to operating power."

"Should that tell us?"

"Sure. As we know, the amount of energy radiated from the sun upon a spot the size of our solar tube is a matter of peanuts compared to the stuff we get out of it. Ergo, our beam must go to Sol and collect the power and draw it back down the beam. Measure the transit-time, and we'll know."

"That's an idea. I've got a micro-clock in the lab. We can measure it to a thousandth of a second. Anyone like to get shook up?"

"How?" asked Jim.

"Snapping from zero to 1-G all to oncet-like isn't too gentle. She'll knock your eyes out."

"Sounds like fun. I'm elected."

"So am I," insisted Christine.

"That's out," said Jim. "I know what he's talking about."

"So do I," said Arden. "Don't do it."

"Well, what better have you to offer?"

"You and I are going down to the Mall."

Channing groaned in mock anguish. "Here goes another closet full of female haberdashery. I'm going to close that corridor some day, or put a ceiling on the quantity of sales, or make it illegal to sell a woman anything unless she can prove that 'she has nothing to wear!'"

"That, I'd like to see," said Walt.

"You would," snorted Arden. "Come on, Chris. Better than the best of three worlds is available."

"That sort of leaves me all alone," said Don. "I'm going to look up Wes Farrell and see if he's been able to make anything worth looking at for a driver detector."

Don found Wes in the laboratory, pouring over a complicated circuit. Farrell was muttering under his breath, and probing deep into the maze of haywire on the bench.

"Wes, when you get to talking to yourself, it's time to take a jaunt to Joe's."

"Not right now," objected Wes. "I haven't got that hollow leg that your gang seem to have developed. Besides, I'm on the trail of something."

"Yes?" Channing forgot about Joe's, and was all interest.

"I got a wiggle out of the meter there a few minutes ago. I'm trying to get another one."

"What was it like?"

"Wavered up and down like fierce for about a minute after I turned it on. Then it died quick, and has been dead ever since."

"Could it have been anything cockeyed with the instruments?"

"Nope. I've checked every part in this circuit, and everything is as good as it ever will be. No, something external caused that response."

"You've tried the solar tube with a dynode of the same alloy as the driver cathodes?"

"Uh-huh. Nothing at all. Oh, I'll take that back. I got a scratch. With a pre-meter gain of about four hundred decibels, I read three micromicroamperes. That was detected from a driver tube forty feet across the room, running at full output. I wondered for a minute whether the opposing driver was doing any cancellation, and so I took a chance and killed it for about a half second, but that wasn't it."

"Nuts. Does the stuff attenuate with distance?"

"As best as I could measure, it was something to the tune of inversely proportional to the cube root of the distance. That's normal for beams of a not-too-tight nature and it shows that the stuff isn't globularly radiated. But the amplifier gain was hanging right on the limit of possible amplification, and the meter was as sensitive as a meter can be made, I think. You couldn't talk from one end of Venus Equilateral to the other with a set like that."

"No, I guess you're right. Hey! Look!"

The meter took a sudden upswing, danced for a minute, and died once more.

"What have you got in there? What did you change?"

"Oh, I got foolish and tried a tuned circuit across the output of one of the miniature transmission tubes. It's far enough away from the big beams and stuff at the North end so that none of the leakage can cause trouble. Besides, I'm not getting anything like our beam transmissions."

Channing laughed. "Uh-huh, looks to me like you're not getting much of anything at all."

Farrell smiled wryly. "Yeah, that's so," he agreed. "But look, Don, Hertz himself didn't collect a transcontinental short-wave broadcast on his first attempt."

"If Hertz had been forced to rely upon vacuum tubes, his theories couldn't have been formulated, I think," said Channing. "At least, not by him. The easier frequencies and wave lengths are too long; a five hundred meter dipole can't be set up in a small room for laboratory tinkering. The kind of frequencies that come of dipoles a couple of feet long, such as Hertz used, are pretty hard to work with unless you have special tubes."

"Hertz had rotten detectors, too. But he made his experiments with spark-gap generators, which gave sufficient high-peak transients to induce spark-magnitude voltages in his receiving dipole."

"I'm not too certain of that tuned-circuit idea of yours, Wes. Go ahead and tinker to your heart's content, but remember that I'm skeptical of the standard resonance idea."

"Why?"

"Because we've been tinkering with driver tubes for years and years—and we have also been gadgeting up detectors, radio hootnannies, and stuff of the electronic spectrum all the way from direct current to hard X rays, and we have yet to have anything react to driver radiation. Ergo, I'm skeptical."

The call bell rang for Channing, and he answered. It was Walt Franks.

"Don," he said with a laugh in his voice, though it was apparent that he felt slightly guilty about laughing, "got a 'gram from Addison, the project engineer on the solar beam from Terran Electric. Says: 'Finally got through Channing Layer. Power by the megawatt-hour in great shape. But the atmosphere from the Channing Layer right down to the snout of the tube is a dull red scintillation. Something like the driver-tube trail—but it ionizes the atmosphere into ozone. Power by the megawatt, and ozone by the megaton.'"

"Ozone, hey? Lots of it?"

"Plenty, according to the rest of this. It looks to me like a sort of 'denatured' power system. There it is, all nice and potent, cheap, and unlicensed. But the second swallow going down meets the first one on the way back. Power they got—but the ozone they can't take; it's poisonous like a nice dose of chlorine. Poor Terran Electric!"

Mark Kingman sat in the control room of a ship of space, and worried. Below the dome, Venus covered three quarters of the sky, and it circled slowly as the Terran Electric ship oscillated gently up and down.

Before Kingman, on the desk, were pages of stock market reports. On a blackboard, a jagged line denoted the vacillation of Interplanetary Communications, Preferred. This phase of his plan was working to perfection. Gradually, he was burning share after share out of uninterested hands, by his depredations. Soon he would have enough stock in Interplanetary to stage a grand show, and then he could swing the thing his way.

His worry was not with this affair.

He gloated over that. His belief that he could beat the Venus Equilateral crowd if he fought them on his ground with his weapon was being

corroborated. That, plus the fact that he was using some of Venus Equilateral's own thunder to do the job, was giving him to think that it was but a matter of time.

And, he laughed, the poor fools were not aware of their peril. Oh, some bird was trying to buck him, but he was not prepared as Kingman was, nor had he the source of information that Kingman had.

No, the thing that worried him was—

And there it came again! A wild, cacophonous wailing, like a whole orchestra of instruments playing at random, in random keys. It shook the very roots of the body, that terrible caterwauling, and not only did it shake the body, and the mind, but it actually caused loose plates to rattle in the bulkhead, and the cabinet doors followed in unison. The diapason stop was out for noon, and the racket filled the small control room and bounced back and forth, dinning at the ears of Kingman as it went echoing by. It penetrated to the upper reaches of the ship, and the crew gritted their teeth and cursed the necessity of being able to hear orders, for cotton plugs would have been a godsend and a curse simultaneously. Anything that would blot that racket out would also deafen them to the vital orders necessary to the operation of the ship in this precarious poisoning maneuver.

Two hundred sheer watts of undistorted audio power boomed forth in that tiny room—two hundred watts of pure, undistorted power to racket forth something that probably started out as sheer distortion.

And yet—

Faintly striving against that fearful racket there came a piping, flat-sounding human voice that said: "Kingman! I.C. Preferred just hit eighty-nine!"

Kingman scowled and punched on the intership teletype machine. Using the communicator set with that racket would have been impossible.

The radio man read the note that appeared on his 'type, and smiled grimly. He saw to his helio-mirror and sighted through a fine telescope at a spot on Venus, three thousand miles below. The helio began to send its flashing signal to this isolated spot near the Boiling River, and it was read, acknowledged, and repeated for safety's sake. The radio man flashed "O.K." and went back to his forty-seventh game of chess with the assistant pilot.



The helio man on the Boiling River read the message, grinned, and stepped to the telephone. He called a number at Northern Landing, and a tight beam sped across the Northern quarter of Venus to a man connected with the Venus Stock Market. The man nodded, and said to another: "Buy fifteen hundred—use the name of Ralph Gantry this time."

The stock purchased under the name of Ralph Gantry was signed, sealed, and delivered exactly fourteen minutes before the ticker projection on the grand wall of the Exchange showed the I.C. Preferred stock turn the bottom curve and start upward by hitting Eighty-nine!

Back in the Terran Electric spaceship, Kingman's ears were still beset by the roaring, alien music.

He was sitting in his chair with his head between his hands, and did not see the man approaching the instrument panel with a pair of side-cutters in one hand. The man reached the panel, lifted it slightly, and reached forward. Then Kingman, hearing a slight imperfection in the wail of the speaker, looked up, jumped from his chair, and tackled the engineer.

"You blasted fool!" blazed Kingman. "You idiot!"

The music stopped at his third word, and the scream of his voice in the silence of the room almost scared Kingman himself.

"Mark, I'm going nuts. I can't stand that racket."

"You're going to stand it. Unless you can get something to cut it out."

"I can't. I'm not brilliant enough to devise a circuit that will cut that noise and still permit the entry of your fellow on Luna."

"Then you'll live with it."

"Mark, why can't we take that relay apart and work on it?"

"Ben, as far as I know, that relay is what Channing and his gang would give their whole Station for—and will, soon enough. I don't care how it works—or why."

"That's no way to make progress," objected Ben.

"Yeah, but we've got the only detector for driver radiation in this part of the universe! I'm not going to have it wrecked by a screwball engineer who doesn't give a care what's going on as long as he can tinker with something new and different. What do we know about it? Nothing. Therefore how can you learn anything about it? What would you look for? What would you expect to find?"

"But where is that music coming from?"

"I don't know. As best as we can calculate, driver radiation propagates at the square of the speed of light, and that gives us a twenty-four minute edge on Venus Equilateral at the present time. For all I know, that music may be

coming from the other end of the galaxy. At the square of the speed of light, you could talk to Centauri and get an answer in not too long."

"But if we had a chance to tinker with that relay, we might be able to find out what tunes it and then we can tune in the Lunar station and tune out that cat-melody."

"I'm running this show—and this relay is going to stay right where it is. I don't care a hoot about the control circuit it breaks; those contacts are set, somehow, so that we can detect driver radiation, and I'm not taking any chances of having it ruined."

"Can't you turn the gain down, at least?"

"Nope. We'd miss the gang at Luna."

The speaker spoke in that faint, flat-toned human voice again. It was easy to see that all that gain was necessary to back up the obviously faint response of Kingman's detector. The speaker said: "Kingman! Addison got power through the Channing Layer!"

That was all for about an hour. Meanwhile, the mewling tones burst forth again and again, assaulting the ears with intent to do damage. The messages were terse and for the most part, interesting. They gave the market reports; they intercepted the beam transmissions through the Terran Heavyside Layer before they got through the Lunar Relay Station, inspected the swiftly-moving tape and transmitted the juicy morsels to Kingman via the big driver tube that stood poised outside of the landed spaceship.

Kingman enjoyed an hour of celebration at Addison's success, and then the joy turned to bitter hate as the message came through telling of the ozone that resulted in the passage of the solar beam through the atmosphere. The success of the beam, and the utter impossibility of using it were far worse than the original fact of the beam's failure to pass the Channing Layer.

So Kingman went back to his stock market machinations, and applied himself diligently. And as the days wore on, Kingman's group manipulated their watered stock and ran the price up and down at will, and after each cycle Kingman's outfit owned just one bit more of Venus Equilateral.

Terran Electric would emerge from this battle with Interplanetary Communications as a subsidiary—with Kingman at the helm!

Walt Franks entered Channing's office with a wild-eyed look on his face.
"Don! C^2 !"

"Huh? What are you driving about?"

" C^2 . The speed of light, squared!"

"Fast—but what is it?"

"The solar beam! It propagates at C^2 !"

"Oh, now look. Nothing can travel that fast!"

"Maybe this isn't *something*!"

"It has energy, energy has mass, mass cannot travel faster than the limiting speed of light."

"O.K. It can't do it. But unless my measurements are all haywire, the beam gets to Sol and back at C^2 . I can prove it."

"Yeah? How? You couldn't possibly measure an interval so small as two times sixty-seven million miles—the radius of Venus' orbit—traversed at the speed of light squared."

"No. I admit that. But, Don, I got power out of Sirius!"

"You WHAT?" yelled Channing.

"Got power out of Sirius. And unless I can't use a micro-clock, it figured out from here to Sirius and back with the bacon in just about ninety-three percent of the speed of light, squared. Seven percent is well within the experimental error, I think, since we think of Sirius as being eight-and-one-half light-years away. That's probably not too accurate as a matter of fact, but it's the figure I used. But here we are. Power from Sirius at C^2 . Thirty-five billion miles per second! This stuff doesn't care how many laws it breaks!"

"Hm-m-m. C^2 , hey? Oh, lovely. Look, Walt, let's run up and take a whirl at Wes Farrell's detector. I'm beginning to envision person-to-person service, ship-to-ship service, and possibly the first Inter-planet Network. Imagine hearing a play-by-play description of the Interplanetary Series!"

"Wool-gathering," snorted Walt. "We've gotta catch our detector first!"

"Wes has something. First glimmer we've had. I think this is the time to rush into it with all feet and start pushing!"

"O.K. Who do we want?"

"Same gang as usual. Charley and Freddy Thomas, Walton, Jim Warren, Wes Farrell, of course, and you can get Jim Baler into it too. No, Walt, Christine Baler is not the kind of people you haul into a screwdriver meeting."

"I was merely thinking."

"I know. But you're needed, and if she were around, you'd be a total loss as far as cerebration."

"I like her."

"So does Barney Carroll."

"Um. Hadn't thought of that one. O.K., no Christine in our conference. I'll have Jeanne call the screwballs on the communicator."

They dribbled into Farrell's laboratory one by one, and then Don said:

"We have a detector. It is about as efficient as a slab of marble; only more so. We can get a tinkle of about ten micromicroamps at twenty feet distance from a driver tube using eight KVA input, which if we rate this in the usual spaceship efficiency, comes to about one-half G. That's about standard, for driver tubes, since they run four to a ship at 2-G total.

"Now, that is peanuts. We should be able to wind a megameter around the peg at twenty feet. Why the red ionization comes out of the tube and hits our so-called detector, and the amount of ozone it creates is terrific. Yet we can't get a good reading out of it."

Walt asked: "Wes, what worked, finally?"

"A four-turn coil on a ceramic form, in series with a twenty micromicrofarad tuning condenser. I've been using a circular plate as a collector."

"Does it tune?"

"Nope. Funny thing, though, it won't work without a condenser in the circuit. I can use anything at all there without tuning it. But, darn it, the coil is the only one that works."

"That's slightly ridiculous. Have you reconstructed all factors?"

"Inductance, distributed capacity, and factor 'Q' are all right on the button with two more I made. Nothing dioding."

"Hm-m-m. This takes the cake. Nothing works, you say?"

"Nothing in my mind. I've tried about three hundred similar coils, and not a wiggle since. That's the only one."

Charley Thomas said: "Wes, have you tried your tube-amplifier system ahead of it?"

"Yes, and nothing at all happens then. I don't understand that one, because we know that any kind of input power will be re-beamed as similar power. I should think that the thing would amplify the same kind of stuff. I've used a solar beam miniature with a driver-alloy dynode in it, but that doesn't work either."

"Shucks," said Charley.

Don stood up and picked up the coil. "Fellows, I'm going to make a grand, old college try!"

"Yes?" asked Walt.

"I've got a grand idea, here. One, I'm still remembering that business of making the receptor dynode of the same alloy as the transmitter cathode. I've a hunch that this thing is not so much an inductor, but something sour in the way of alloy-selectivity. If I'm right, I may cut this in half, and make two detectors, each of similar characteristics. Shall I?"

"Go ahead. We've established the fact that it is not the physico-electrical characteristics of that coil," said Wes. "I, too, took my chances and rewound that same wire on a couple of other forms. So it doesn't count as far as an inductance goes. So we can't ruin anything but the total make-up of the wire. I think we may be able to re-establish the wire by self-welding if your idea doesn't work. Now, unless we want to search the three planets for another hunk of wire to work like this one did, without knowing what to look for and therefore trying every foot of wire on three planets—"

"I'll cut it," said Channing with a smile. His cutters snipped, and then fastened one end of the wire to the coil, stripping the other portion off and handing it to Charley Thomas, who rewound it on another form.

"Now," said Don, "crank up your outfit and we'll try this hunk."

The beam tubes were fired up, and the smell of ozone began to make itself prominent. Channing cranked up the air-vent capacity to remove the ozone more swiftly. The men applied themselves to the detector circuits, and Wes, who recognized the results, said: "This hunk works. About as good as the whole coil."

Channing replaced the first coil with the second. Wes inspected the results and said: "Not quite as good, but it does work."

Walt nodded, and said: "Maybe it should be incandescent."

"That's a thought. Our solar beam uses an incandescent dynode." Channing removed the second coil and handed it to Freddy. "Take this thing down to the metallurgical lab and tell 'em to analyze it right down to the trace of sodium that seems to be in everything. I want quantitative figures on every element in it. Also, cut off a hunk and see if the crystallographic expert can detect anything peculiar, that would make this hunk of copper wire different from any other hunk. Follow?"

"Yup," said Freddy. "We'll also start making similar alloys with a few percent variation on the composition metals. Right?"

"That's the ticket. Wes, can we evacuate a tube with this wire in it and make it incandescent?"

"Let's evacuate the room. I like that stunt."

"You're the engineer on this trick. Do it your way."

"Thanks. I get the program, all right. Why not have Charley build us a modulator for the driver tube? Then when we get this thing perfected, we'll have some way to test it."

"Can do, Charley?"

"I think so. It's easy. We'll just modulate the cathode current of the electron guns that bombard the big cathode. That is the way we adjust for drive; it should work as a means of amplitude-modulation."

"O.K.," said Channing. "We're on the rails for this one. We'll get together as soon as our various laboratories have their answers and have something further to work with."

Above Venus, Mark Kingman was listening to the wailing roar of alien symphony and cursing because he could hardly hear the voice of his Lunar accomplice saying: "V. E. Preferred just hit one hundred and two!"

Fifteen minutes before the peak hit Northern Landing, share after share was being dumped, and in addition, a message was on its way back to Terra. It went on the regular beam transmission through Venus Equilateral, carefully coded. It said:

"Have sufficient stock and additional collateral to ply the first pressure. Apply phase two of plan. Kingman."

In the ten hours that followed, Venus Equilateral stock went down and down and down, passed through a deep valley, and started up again. Up it went. Up past the one hundred mark, up into the one hundred and fifties. It hovered there for a bit, and then started up again. Kingman's crowd was offering twice the market for the preferred stock, and there was little to have. It took a short-time dip at three hundred, and the few minutes of decline smoked a lot of stock out of the hands of people who looked upon this chance as the right time to make their money and get out.

Then the stock began to climb again, and those people who thought that the price had been at its peak-and-passed were angrily trying to buy in again. That accelerated the climb, but Kingman's crowd, operating on Venus and on Mars and on Terra were buying only, and selling not one share of Interplanetary Communications.

Terran Electric stock took a gradual slide, for Kingman's crowd needed additional money. But the slide was slow, and controlled, and manipulated only for the purpose of selling short. Terran Electric stock eventually remained in the hands of Kingman's crowd, though its value was lessened.

I. C. Preferred hit four hundred and sixty-eight, and hovered. It vacillated around that point for another hour, and the market closed at four hundred

and sixty-nine and three-eighths.

Kingman looked at his watch and smiled. He reached forth and cut the dinning sound of the cacophony with a vicious twist of the gain knob. Silence reigned in the spaceship; grand, peaceful silence. Kingman, his nerves frayed by the mental activity and the brain-addling music-from-nowhere, took a hot shower and went to bed.

He locked the panel of the control room first, however. He wanted no engineer tinkering with his pet relay.

Cartright came into Channing's living room with a long face. "It's bad," he said. "Bad."

"What's bad?"

"Oh I, like the rest of the fools, got caught in his trap."

"Whose trap?"

"The wild man who is trying to rock Interplanetary Communications on its axis."

"Well, how?"

"They started to buy like mad, and I held out. Then the thing dropped a few points, and I tried to take a bit of profit, so that we could go on bolstering the market. They grabbed off my stock, and then, just like *that!* the market was on the way up again and I couldn't find more than a few odd shares to buy back."

"Don't worry," said Channing, "I don't think anybody is big enough to really damage us. Someone is playing fast and loose, making a killing. When this is over, we'll still be in business."

"I know, Don, but whose business will it be? Ours, or theirs?"

"Is it that bad?"

"I'm afraid so. One more flurry like today, and they'll be able to tow Venus Equilateral out and make Mars Equilateral out of it, and we won't be able to say a word."

"Hm-m-m. You aren't beaten?"

"Not until the last drop. I'm not bragging when I say that I'm as good an operator as the next. My trouble today was not being a mind reader. I'd been doing all right, so far. I've been letting them ride it up and down with little opposition, and taking off a few here and there as I rode along. Guessing their purpose, I could count on their next move. But this banging the market sky-high has me stumped, or had me stumped for just long enough for me to throw our shirt into the ring. They took that quick—our shirt, I mean."

"That's too bad. What are you leading up to?"

"There are a lot of unstable stocks that a guy could really play hob with; therefore their only reason for picking on I. C. is to gain control!"

"Pirates?"

"Something like that."

"Well," said Channing in a resigned voice, "about all we can do is to do our best and hope we are smart enough to outguess 'em. That's your job, Cartright. A long time ago I. C. made their decision concerning the executive branch of this company, and they elected to run the joint with technical men. The business aspects and all are under the control of men who know what they're fighting. We hire business men, just like business men hire engineers, and for the opposite purpose. You're the best we could get, you know that. If those guys get Venus Equilateral, they'll get you too. But if you do your best and fail, we can't shoot you in the back for it. We'll all go down together. So keep pitching, and remember that we're behind you all the way!"

"Can we float a bit of a loan?"

"Sure, if it's needed. I'd prefer Interplanetary Transport if they'll do business with us. We've been in the way of helping them out a couple of million dollar losses; they might be anxious to reciprocate."

"O.K., I have your power of attorney anyway. If I get in a real crack, I'll scream for I. T. to help. Right?"

"Right."

Cartright left, and as he closed the door, Channing's face took on a deep, long look. He was worried. He put his head between his hands and thought himself into a tight circle from which he could not escape. He did not hear Walt Franks enter behind Arden and Christine.

"Hey!" said Walt. "Why the gloom? I bear glad tidings!"

Channing looked up. "Spill," he said with a glum smile. "I could use some glad tidings right now."

"The lab just reported that that hunk of copper wire was impure. Got a couple of traces of other metals in it. They've been concocting other samples with more and less of the impurities, and Wes has been trying them as they were ready. We've got the detector working to the point where Freddy has taken the *Relay Girl* out for a run around the Station at about five hundred miles and Wes is still getting responses!"

"Is he? How can he know?"

"Charley rigged the *Relay Girl's* drivers up with a voice modulator, and Freddy is jerking his head off because the acceleration is directly proportional to the amplitude of his voice, saying: 'One, two, three, four, test.' Don, have you ever wondered why an engineer can't count above four?"

"Walt, does it take a lot of soup to modulate a driver?" asked Arden.

"Peanuts," grinned Franks. "This stuff is not like the good old radio; the power for driving the spaceship is derived mostly from the total disintegration of the cathode and the voltage applied to the various electrodes is merely for the purpose of setting up the proper field-conditions. They draw quite a bit of current, but nothing like that which would be required to lift a spaceship at 2-G for a hundred hours flat."

He turned back to Channing and said: "What's the gloom?"

Don smiled in a thoughtful fashion. "It doesn't look so bad right now. Some gang of stock market cutthroats have been playing football with Interplanetary Communications, and Cartright says he is sure that they want control. It's bad; he's been clipped a couple of hard licks, but we're still pitching. The thing I'm wondering right now is this: Shall we toss this possibility of person-to-person and ship-to-ship just at the right turn of the

market to bollix up their machinations, or shall we keep it to ourselves and start up another company with this as our basis?"

"Can we screw 'em up by announcing it?"

"Sure. If we drop this idea just at the time they're trying to run the stock down, it'll cross over and take a run up, which will set 'em on their ear."

"I don't know. Better keep it to ourselves for a bit. Something may turn up. But come on down to Wes' lab and give a look at our new set-up."

Channing stood up and stretched. "I'm on the way," he said.

Farrell was working furiously on the detector device, and as they entered, he indicated the meter that was jumping up and down. Out of a speaker there was coming the full, rich tones of Freddy Thomas' voice, announcing solemnly: "One, two, three, four, test."

Wes said: "I'm getting better. Charley has been bettering his modulator now, and the detector is three notches closer to whatever this level of energy uses for resonance. Evacuation and the subsequent incandescence was the answer. Another thing I've found is this—" Farrell held up a flat disk about six inches in diameter with a sawcut from edge to center. "As you see, the color of this disk changes from this edge of the cut, varying all the way around the disk to the other side of the cut. The darned disk is a varying alloy—I've discovered how to tune the driver radiation through a limited range. We hit resonance of the *Relay Girl's* driver system just off the end of this disk. But watch while I turn the one in the set."

Farrell took a large knob and turned it. Freddy's voice faded, and became toneless. Farrell returned the knob to its original position and the reception cleared again. "Inside of that tube there," said Farrell, "I have a selsyn turning the disk, and a small induction loop that heats the whole disk to incandescence. A brush makes contact with the edge of the disk and the axle makes the center connection. Apparently this stuff passes on a direct line right through the metal, for it works."

"Have you tried any kind of tube amplification?" asked Don.

"Not yet. Shall we?"

"Why not? I can still think that the relay tube will amplify if we hook up the input and output loads correctly."

"I've got a tube already hooked up," said Walt. "It's mounted in a panel with the proper voltage supplies and so on. If your resistance calculation is correct, we should get about three thousand voltage gain out of it."

He left, and returned in a few minutes with the tube. They busied themselves with the connections, and then Don applied the power.

Nothing happened.

"Run a line from the output back through a voltage-dividing circuit to the in-phase anode," suggested Walt.

"How much?"

"Put a potentiometer in it so we can vary the amount of voltage. After all, Barney Carroll said that the application of voltage in phase with the transmitted power is necessary to the operation of the relay tube. In transmission of D. C., it is necessary to jack up the in-phase anode with a bit of D. C. That's in-phase with a vengeance!"

"What you're thinking is that whatever this sub-level energy is, some of it should be applied to the in-phase anode?"

"Nothing but."

The cabinet provided a standard potentiometer, and as Don advanced the amount of fed-back voltage, Freddy's voice came booming in louder and louder. It overloaded the audio amplifier, and they turned the gain down as Channing increased the in-phase voltage more and more. It passed through a peak, and then Don left the potentiometer for maximum.

"Wes," he said, "call Freddy and tell him to take off for Terra, at about 4-G. Have the gang upstairs hang a ship beam on him so we can follow him with suggestions. Too bad we can't get there immediately."

"What I'm worrying about is the available gain," said Wes. "That thing may have given us a gain of a couple of thousand, but that isn't going to be enough. Not for planet-to-planet service."

"Later on we may be able to hang a couple of those things in cascade," suggested Walt.

"Or if not, I know a trick that will work—one that will enable us to get a gain of several million."

"Yeah? Mirrors, or adding machines? You can't make an audio amplifier of a three million gain."

"I know it—at least not a practical one. But, we can probably use our audio modulator to modulate a radio frequency, and then modulate the driver with the RF. Then we hang a receiver on to the detector gadget here, and collect RF, modulated, just like a standard radio transmission, and amplify it at RF, convert it to IF, and detect it to AF. Catch?"

"Sure. And that gives me another thought. It might just be possible, if your idea is possible, that we can insert several frequencies of RF into the tube and hang a number of receivers on the detector, here."

Arden laughed. "From crystal detection to multiplex transmission in ten easy lessons."

"Call Charley and have him begin to concoct an RF stage for tube-modulation," said Don. "It'll have to be fairly low—not higher than a couple of megacycles so that he can handle it with the stuff he has available, but as long as we can hear his dulcet voice chirping that 'one, two, three, four, test,' of his, we can also have ship-to-Station two-way. We squirt out on the ship beam, and he talks back on the driver transmitter."

"That'll be a help," observed Wes. "I'd been thinking by habit that we had no way to get word back from the *Relay Girl*."

"So had I," confessed Walt. "But we'll get over that."

"Meanwhile, I'm going to get this alloy-selectivity investigated right down to the last nub," said Don. "Charley's gang can take it from all angles and record their findings. We'll ultimately be able to devise a system of mathematics for it from their analysis. You won't mind being bothered every fifteen minutes for the first week, will you, Wes? They'll be running to you

in your sleep with questions until they catch up with your present level of ability in this job. Eventually they'll pass you up, and then you'll have to study their results in order to keep up."

"Suits me. That sounds like my job anyway."

"It is. O.K., Arden, I'm coming now."

"It's about time," smiled Arden. "I wouldn't haul you away from your first love excepting that I know you haven't eaten in eight or nine hours. I've got roast knolla."

"S'long, fellows," grinned Channing. "I'm one of the few guys in the inner system who can forget that the knolla is the North Venus brother to a pussy cat."

"I could feed you pussy cat and you'd eat it if I called it knolla," said Arden. "But you wouldn't eat knolla if I called it pussy cat."

"You can't tell the difference," said Walt.

"Tell me," asked Wes, "what does pussy cat taste like?"

"I mean by visual inspection. Unfortunately, there can be no comparison drawn. The Venusians will eat pussy cat, but they look upon the knolla as a household pet, not fit for Venusian consumption. So unless we revive one of the Ancient Martians, who may have the intestinal fortitude—better known as guts—to eat both and describe the difference, we may never know," offered Walt.

"Stop it," said Arden, "or you'll have my dinner spoiled for me."

"All the more for me," said Don. "Now, when I was in college, we cooked the dean's cat and offered it to some pledges under the name of knolla. They said—"

"We'll have macaroni for dinner," said Arden firmly. "I'll never be able to look a fried knolla in the pan again without wondering whether it caterwauled on some back fence in Chicago, or a Palanortis Whitewood on Venus."

She left, and Channing went with her, arguing as he went to the effect that she should develop a disregard for things like their discussion. As a matter of

interest, Channing had his roast knolla that evening, so he must have convinced Arden.

Walt said: "And then there were three. Christine, has our little pre-dinner talk disturbed your appetite?"

"Not in the least," said the girl stoutly. "I wouldn't care whether it was knolla or pussy cat. I've been on Mars so long that either one of the little felines is alien to me. What have you to offer?"

"We'll hit Joe's for dinner, which is the best bar in sixty million miles today. Later we may take in the latest celluloid epic, and then there will be a bit of mixed wrestling in the ballroom."

"Mixed wres—Oh, you mean dancing. Sounds interesting. Now?"

"Now. Wes, what are you heading for?"

"Oh, I've got on a cockeyed schedule," said Wes. "I've been catching my sleep at more and more out-of-phase hours until this is not too long after breakfast for me. You birds all speak of 'Tomorrow,' 'Today' and 'Yesterday' out here, but this business of having no sun to come up in the morning, and the electric lights running all the time has me all bollixed up."

"That daily nomenclature is purely from habit," said Walt. "As you know, we run three equal shifts of eight hours each, and therefore what may be 'Morning' to Bill is 'Noon' to James and 'Night' to Harry. It is meaningless, but habitual to speak of 'Morning' when you mean 'Just after I get up!' Follow me?"

"Yup. This, then, is morning to me. Run along and have fun."

"We'll try," said Walt.

"We will," said Christine.

Farrell grinned as they left. He looked at Walt, and said: "You will!"

Walt wondered whether he should have questioned Wes about that remark, but he did not. Several hours later, Walt wondered how Wes could have been so right.

Interplanetary Communications, Preferred, started in its long climb as soon as the markets opened, on the following day. Cartright, following his orders and his experience, held onto whatever stock he had, and bought whatever stock was tossed his way. Several times he was on the verge of asking Interplanetary Transport for monetary assistance, but the real need never materialized.

Kingman alternately cursed the whining music and cheered the pyramiding stock. About the only thing that kept Kingman from going completely mad was the fact that the alien music was not continuous, but it came and went in stretches of anything from five to fifty minutes, with varied periods of silence in between selections.

Up and up it went, and Kingman was seeing the final, victorious coup in the offing. A week more, and Venus Equilateral would belong to Terran Electric. The beam from Terra was silent, save for a few items of interest not connected with the market. Kingman's men were given the latest news, baseball scores, and so forth, among which items was another message to Channing from the solar beam project engineer, Addison. They had about given up. Nothing they could do would prevent the formation of ozone by the ton as they drew power by the kilowatt from Sol.

On Venus Equilateral, Channing said: "Ask Charley what his radio frequency is."

Ten minutes later, at the speed of light, the ship beam reached the *Relay Girl* and the message clicked out. Charley Thomas read it, and spoke into the microphone. The *Relay Girl* bucked unmercifully, as the voice amplitude made the acceleration change. Then at the speed of light, squared, the answer came back in less than a twinkle.

"Seventeen hundred kilocycles."

Channing began to turn the tuner of the radio receiver. The band was dead, and Channing laughed. "This is going to be tricky, what with the necessity of aligning both the driver-alloy disk and the radio receiver. Takes time."

He changed the alloy disk in minute increments, and waved the tuner across that portion of the band that would most likely cover the experimental error of Charley Thomas' frequency measurement. A burst of sound caught his ear,

was lost for a moment, and then swelled into perfect tune as Don worked over the double tuning system.

"Whoa, Tillie," said Walt. "That sounds like—"

"Like hell."

"Right. Just what I was going to say. Is it music?"

"Could be. I've got a slightly tin ear, you know."

"Mine is fair," said Walt, "but it might as well be solid brass as far as this mess is concerned. It's music of some kind, you can tell it by the rhythm. But the scale isn't anything like I've heard before."

"Might be a phonograph record played backward," suggested Wes.

"I doubt it," said Channing seriously. "The swell of that orchestra indicates a number of instruments—of some cockeyed kind or other—the point I'm making is that anything of a classical or semi-classical nature played backwards on a phonograph actually sounds passable. I can't say the same for jamstead music, but it holds for most of the classics, believe it or not. This sounds strictly from hunger."

"Or hatred. Maybe the musicians do not like one another."

"Then they should lambaste one another with their instruments, not paste the sub-ether with 'em."

Channing lit a cigarette. "Mark the dial," he said. "Both of 'em. I've got to get in touch with the Thomas Boys."

Walt marked the dials and tuned for the *Relay Girl*. He found it coming in not far from the other setting. Charley was speaking, and they tuned in near the middle of his speech.

"—this thing so that it will not buck like a scenic railway finding the fourth derivative of space with respect to time. For my nontechnical listeners, that is none other than the better known term: Jerkiness. We applied the modulation in to the first driver anode—the little circular one right above the cathode. I don't know whether this is getting out as it should, so I'm going to talk along for the next fifteen minutes straight until I hear from you. Then we're switching over and repeating. Can you hear me?"

Channing cut the gain down to a whisper and put a message on the beam, confirming his reception. Ten minutes later, Charley changed his set speech, and said: "Good! Too bad we haven't got one of those receivers here, or we could make this a two-way with some action. Now listen, Don. My idiot brother says he can make the beam transmit without the drive. Unfortunately, I am not a driver expert like he is and so I can not remonstrate with the half-wit. So, and right now, we're cutting the supply voltage to the final focusing anode. Whoops! I just floated off the floor and the mike cable is all tangled up in my feet. This free stuff is not as simple as the old fiction writers claimed it was. Things are floating all over the place like mad. The accelerometer says exactly zero, and so you tell me if we are getting out. We're going back on 1-G so that we can sit down again. That's better! Though the idiot—it's a shame to be forced to admit that one of your family is half-witted—didn't wait until we were in position to fall. I almost landed on my head—which is where he was dropped as an infant. How was it? Did you hear my manly voice whilst we were going free? Say 'No' so that my idiot brother will not have anything to say about his brilliant mind. I'm out of breath and we're going back on that home recording of Freddy saying, and I will let him quote, via acetate."



The sound of a phonograph pickup being dropped on a record preceded Freddy's voice saying: "One, two, three, four, test. One—"

Channing cut the gain again. "That red-hot. I thought he was talking all this time."

"Not the Thomas Boys. That comes under the classification of 'Work' which they shun unless they can not get any kind of machine to do it for them," laughed Walt.

Walt turned the dials back to the unearthly symphony. "At C^2 , that might come from Sirius," he said, listening carefully. "Sounds like Chinese."

"Oh, now look," objected Don. "What off earth would a Chinese Symphony be doing with a driver-modulator system?"

"Broadcasting—"

"Nope. The idea of detecting driver radiation is as old as the hills. If any culture had uncovered driver-beam transmission, we'd all have been aware of it. So far as I know, we, and the Terran Electric crowd are the only ones who have had any kind of an opportunity of working with this sub-etheric energy. Wes, have you another miniature of the relay tube handy?"

"Sure. Why?"

"I'm going to see if this stuff can be made directional. You're bringing whatever it is into the place on a collector plate and slamming it into an input-terminal power-transmission tube. It goes across the table to the relay tube, and is amplified, and then is tossed across more table to the load-terminal tube, where the output is impressed across your alloy disk. Right?"

"Right."

"I want another relay tube. I'm going to use it for a directional input-beam, aligning it in the same way that Jim Baler and Barney Carroll did their first find. The one that sucked power out of the electric light, turned off the city hall, and so on. Follow?"

"Perfectly. Yes, I've got a couple of them. But they're not connected like Walt's set-up was."

"Well, that three-tube system was built on sheer guesswork some time ago. We can tap in the relay tube and haul out a set of cables that will energize the first relay tube. Hang her on gymbals, and we'll go hunting."

"Shall I have Freddy return?"

"Yes. We'll have Walton's gang build us up about six of these things just as we have here."

"That won't take long," said Walt. "They're working on the tuning disks now, and we should have 'em by the time that Freddy gets back here."

"But this wild and woolly music. It's alien!"

Wes turned from the teletype and dug in the cabinet for the extra relay tube. He up-ended the chassis containing Walt's set-up, and began to attach leads to the voltage supply, cabling them neatly and in accordance with the restrictions on lead-capacities that some of the anodes needed.

"It's alien," said Wes in agreement. "I'm going to shut it off now whilst I tinker with the tube."

"Wait a minute," said Don. "Here comes Jim. Maybe he'd like to hear it."

"Hear what?" asked Jim Baler, entering the door.

"We've a Sirian Symphony," explained Don, giving Jim the background all the way up to the present time. Jim listened, and then said:

"As an engineer, I've never heard anything like that in my life before. But, as a student of ancient languages and arts and sciences, I have. That's Chinese."

"Oh no!"

"Oh yes. But definitely."

"Ye gods!"

"I agree."

"But how—where—"

"And/or when?"

Channing sat down hard. He stared at the wall for minutes. "Chinese. Oh, great, slippery, green, howling catfish!" He picked up the phone and called the decoupler room where the messages were sorted as to destination upon their entry into the Station.

"Ben? Look, have we a ship beam on anything of Chinese registry?"

Ben said wait a minute while he checked. He returned and said: "Four. *The Lady of Cathay*, *The Mandarin's Daughter*, *The Dragoness*, and *The Mongol Maid*. Why?"

"Put a ship message on each of 'em, asking whether they have any Chinese music aboard."

"And then what? They can't answer."

"Make this an experimental request. If any of them are using any recordings of Chinese music, tell them to have their electronics chief replace the phonograph pickup with a microphone—disturbing absolutely nothing—and to reply as if we could hear them. Get me?"

"Can you? Hear 'em, I mean."

"We hear something, and Jim says it's Chinese."

"It's worth a try, then. See you later."

"Will they?" asked Jim, interested in the workings of this idea.

"Sure. Ever since we steered the *Empress of Kolain* out of the grease with the first Station-to-ship beam, all three of the interplanetary companies have been more than willing to co-operate with any of our requests as long as we precede the message with the explanation that it is experimental. They'll do anything we ask 'em to, short of scuttling the ship."

"Nice hookup. Hope it works."

"So do I," said Wes. "This, I mean. I've got our directional gadget hooked up."

"Turn it on."

The wailing of the music came in strong and clear. Wes turned the input tube on its support, and the music passed through a loud peak and died off on the far side to almost zero. Wes adjusted the mobile tube for maximum response and tightened a small set-screw. "It's a shame we haven't got a nice set of protractors and gymbals," said Wes. "I had to tear into the desk lamp to get that flexible pipe."

"Small loss. She's directional, all right. We'll get the gymbals later. Right now I don't want this turned off because we may hear something interesting—Whoops, it went off by itself!"

"Could we dare to hope?" asked Walt.

"Let's wait. They'll have to hitch the microphone on—"

"Give 'em a half hour, at least."

Twenty minutes later, a strange voice came through the speaker. "Dr. Channing, of Venus Equilateral? We have been contacted by your organization with respect to the possibility of your being able to hear the intership communicator system. This seems impossible, but we are not ones to question. The fact that you are in possession of the facts concerning our

love of the music of our ancestors is proof enough that you must have heard something. I presume that further information is desired, and I shall wait for your return. This is Ling Kai Chaing, Captain of the *Lady of Cathay*."

"We got it!" chortled Don. He did a war dance in the lab, and the rest followed suit. Bits of wire and oddments of one sort or another filled the air as the big, grown-up men did a spring dance and strewed the floor with daintily thrown junk. At the height of the racket, Arden and Christine entered—no, they were literally hauled in, completely surrounded, and almost smothered.

Arden fought herself free and said: "What's going on?"

"We've just contacted a ship in space."

"So what? Haven't we been doing that for months?"

"They've just contacted us, too!"

"Huh?" asked Arden, her eyes widening.

"None other. Wait, I'll get an answer." Don contacted Ben, in the decoupler room and said: "Ben, hang this line on the *Lady of Cathay's* beam, will you?"

"Is that her?"

"None other."

"Go ahead. She's coupled."

Don pecked out a message. "Please describe the intercommunication system used by your ship in detail. We have heard you, and you are, therefore, the first ship to contact Venus Equilateral from space flight. Congratulations."

Eight minutes later, the voice of Captain Chaing returned.

"Dr. Channing, I am handing the microphone over to Ling Wey, our electronics engineer, who knows the system in and out. He'll work with you on this problem."

Ling Wey said: "Hello. This is great. But I'm not certain of how it's done. The output of the phono system is very small, and certainly not capable of putting out the power necessary to reach Venus Equilateral from here. However, we are using a wired-radio system at seventeen hundred and

ninety kilocycles in lieu of the usual cable system. The crew all like music, and, therefore, we play the recordings of our ancestral musicians almost incessantly."

He paused for breath, and Channing said: "Walt, tap out a message concerning the lead-length of the cables that supply the driver anodes. Have him check them for radio frequency pickup."

"I get it." The 'type began to click.

This communication was carried on for hour after hour. Don's guess was right, it turned out; the lead that connected the first driver anode was tuned in wave length to almost perfect resonance with the frequency of the wired-radio communicator system. Channing thanked them profusely, and they rang off. Soon afterward the wailing, moaning music returned to the air.

"Wonder if we could get that without the radio," said Don.

"Don't know. We can pack the juice on in the amplifier and see, now that we have it tuned on the button," said Walt.

"It won't," said Wes. "I've been all across the dial of the alloy disk. Nothing at all."

"O.K. Well, so what if it doesn't. We've still got us a ship-to-ship communications system. Hey! What was that?"

That was a pale, flat-sounding human voice saying: "Kingman! I. C. Pfd. has been at six hundred and nine for two days, now. What's our next move?"

"Kingman!" exploded Channing. "Why, the ... the—"

"Careful," warned Arden. "There's a lady present."

"Huh?"

"Her," said Arden pointing at Christine.

"Wait," said Walt. "Maybe he'll answer."

Don fiddled with the dials for a full fifteen minutes, keeping them very close to the spot marked, hoping that Kingman's answer might not be too far out of

tune. He gave up as the answer was not to be found, and returned to the original setting. Ten minutes later the voice said: "Kingman, where in the devil is my answer? I want to know what our next move is. There isn't a bit of V. E. stock available. Why don't you answer?"

Then, dimly in the background, a voice spoke to the operator of the instrument. "Kingman's probably asleep. That terrible moaning-stuff he's been complaining about makes him turn the thing off as soon as the day's market is off. He—and the rest of that crew—can't stand it. You'll have to wait until tomorrow's market opens before he'll be listening."

"O.K.," said the operator, and the set went silent.

"Kingman!" said Don Channing in a low, hard voice. "So he's the bright guy behind this. I get it now. Somehow he discovered a detector, and he's been playing the market by getting the quotations by sub-etheric transmission at C^2 and beating the Northern Landing market. And did you get the latest bit of luck? Kingman still is unaware of the fact that we are onto him—and have perfected this C^2 transmission. Here's where he gets caught in his own trap!"

"How?"

"We're not in too bad shape for making good, honest two-ways out of this sub-ether stuff. Kingman is still behind because he hasn't got a return line back to Terra—he must be using our beams, which gives us a return edge."

"Why not get him tossed into the clink?" asked Walt.

"That's practical. Besides, we're sitting in a great big pile of gravy right now. We can prove Kingman has been violating the law to embezzle, mulct, steal, commit grand larceny, and so on. We're going to take a swing at Mr. Kingman and at Terran Electric that they won't forget. We can't lose, because I'm not a good sportsman when I find that I've been tricked. We're going after Kingman in our own fashion—and if we lose, we're going to go tinhorn and cry for the gendarmes. I'm not proud."

"What do you plan?"

"We'll put a horde of folks on the decoupler files with the code Terran Electric filed with the government office. We can get the code, and I'm of the opinion that Kingman wouldn't take time to figure out a new code, so he'll be using the old one. As soon as we find a message in that code that is either

addressed to Terran Electric or pertains to I. C. Preferred stock, we'll start to intercept all such messages and use 'em for our own good."

"That's illegal."

"Yup. But who's gonna holler? Kingman can't."

"But suppose we lose—?"

"Kingman will not know we've been tricking him. Besides, we can't lose with two ways to get ahead of his one. Come on, fellows, we've got to help get the extra receivers together."

"How are we going to cut through the Channing Layer?"

"Easy. That's where we'll use the relay stations at Luna, Deimos, and the six portables that circle Venus."

"I get it. O.K., Don, let's get to work."

"Right. And we'd better leave a guy here to collect any more interesting messages from Kingman's crowd. We can tune it right on to Kingman's alloy, and that'll make that music take the back seat. We need narrower selectivity."

"Charley's gang will find that if it is to be found," smiled Walt. "We're really on the track this time."

A dead-black spaceship drifted across the face of Luna slowly, and its course, though apparently aimless, was the course of a ship or a man hunting something. It darted swiftly, poised, and then zigzagged forward, each straight-side of the jagged course shorter than the one before. It passed over a small crater and stopped short.



Below, there was a spaceship parked beside a driver tube anchored in the pumice.

The black ship hovered above the parked ship, and then dropped sharply, ramming the observation dome on top with its harder, smaller bottom. The two ships tilted and fell, crushing the ground near the poised driver tube. Space-suited men assaulted the damaged ship, broke into the bent and battered plates and emerged with three men who were still struggling to get their suits adjusted properly.

Channing's men took over the poised driver tube, and in their own ship, Walt spoke over a sub-ether radio of a different type.

"Don, we got him."

Don answered from Venus Equilateral, and his voice had no more delay than if he had been within a hundred yards of the crater on Luna.

"Good. Stay there; you can contact the Lunar Relay Station from there. Wes is all ready on Station 3 above Northern Landing with his set, and Jim Baler is at the Deimos Station."

"Hi, Walt," came Wes' voice.

"Hi," said Jim Baler.

"Hello, fellows," said Walt. "Well, what cooks?"

"Kingman," said Channing. "You've got your orders, Walt. When Kingman expects the market to go down, tell him it's still going up. We'll figure this out as we go along, but he won't like it at all."

There was silence for a few minutes, and then Don said: "Walt, Kingman's sent a message through the Northern Landing station now. He says: 'Dump a block to shake the suckers loose. This is pyramided so high that they should all climb on the sell-wagon; running the market down of their own weight. When it hits a new low, we'll buy, and this time end up by having control.' When he starts to run the market down, you buy at Terra."

Minutes later, the message hit the Terra market, and Kingman's agent started to unload. The stock started off at six hundred and nine, and it soon dropped to five-forty. It hovered there, and then took another gradual slide to four-seventy. Then a message came through the regular beam station which Walt intercepted, decoded with Terran Electric's own code book, and read as follows:

"I. C. Preferred coming in fast. Shall we wait?"

Walt chuckled and spoke into the driver modulator. "Kingman," he said, "some wiseacre is still buying. I. C. Preferred is running at seven-ninety! What now?"

In the Venus Equilateral radio, he said: "Don, I just fixed him."

From Venus, Wes said: "You sure did. He's just giving orders to drop some more stock. This is too dirty to be funny, but Kingman asked for it. I know him. He's got this set up so that no one can do a thing on this market program without orders from him. Too bad we can't withhold the Northern Landing quotations from him."

The Lunar Beam brought forth another message intended for Kingman's interceptor at Luna. "I. C. Preferred is dropping like a plummet. When can we buy?"

Walt smiled and said into Kingman's set-up: "Kingman! I. C. Preferred is now at eight hundred and seventy!"

Not many minutes later, Wes said: "That was foul, Walt. He's just given orders to run the market down at any cost."

"O.K.," said Walt. "But he's going to go nuts when the Northern Landing Exchange starts down without ever getting to that mythical nine hundred."

"Let him wonder. Meanwhile, fellows, let's run ourselves a slide on Terran Electric. Sell the works!"

Terran Electric started down just as I. C. Preferred took its third drop. It passed three hundred, and started down the two hundred numbers. Walt shook his head and said to Kingman: "Kingman, we're getting results now. She's dropped back again—to six hundred and three." Then he said: "Kingman, someone is playing hob with T. E. Preferred. She's up to two hundred and fifty-one."

To Don, Walt said: "Good thing that Kingman has that Sinese Chimphony for a bit of mood music, or he'd recognize my voice."

"Which way will he jump?" laughed Don. "That was a slick bit of Kingman-baiting, Walt, in spite of your voice."

"Kingman's taking it hard," said Wes. "We says to drop some of his own stock so that they can use the money to manipulate the I. C. stuff."

"O.K.," said Jim Baler. "This looks like a good time to think about buying some of Kingman's stuff. Right?"

"Wait until his sales hit bottom," said Don. "Walt, tip us off."

"O.K. What now?"

"Wait a bit and see."

Terran Electric went down some more, and then Jim said: "Now?"

"Now," answered Don. "You too, Wes."

"Me too?" asked Walt.

"You continue to sell!"

"Oh-oh," said Wes. "Kingman is wild. He wants to know what's the matter with the market."

"Tell him that your end is all right, and that I. C. Preferred is still going down, but steady."

"O.K.," said Walt.

The hours went by, and Kingman became more and more frantic. I. C. Preferred would be reported at five hundred, but the Northern Landing Exchange said two-ten. Meanwhile, Terran Electric—

"Oh, lovely," said Don. "Beautiful. We've got us a reciprocating market now, better than Kingman's. When she's up at Terra, they're down at Canalopsis and Northern Landing—and vice versa. Keep it pumping, boys, and we'll get enough money to buy Kingman out."

The vacillating market went on, and Don's gang continued to rock the Terran Electric stock. Then as the market was about to close for the day, Don said: "Sell 'em short!"

Terran Electric stock appeared on the market in great quantities. Its value dropped down and down and down, and Kingman, appraised of the fall by Walt, who magnified it by not less than two to one, apparently got frantic again, for he said:

"We're running short. Drop your Terran stock to bolster the I. C. job!"

"Oh, lovely," said Don.

"You said that."

"I repeat it. Look, fellows, gather all the T. E. Preferred and I. C. Preferred you can. Walt, tell him that Terran Electric is dropping fast, so he'll scuttle more of his stuff, and we'll pick it up slowly enough so that we won't raise the market. How're we fixed for I. C. Preferred?"

"Not too bad. Can we hit him once more?"

"Go ahead," said Don.

"Kingman," said Walt. "Kingman! Hell's loose. The Interplanetary Bureau of Criminal Investigations has just decided to look into the Interplanetary Communications angle. They want to know who's trying to grab control of a public carrier!"

Minutes later, Wes said: "Oh, Brother Myrtle. That did it. He just gave orders to drop the whole thing short!"

"Wait until I. C. Preferred hits a new low and then we'll buy," said Don.

The flurry dropped I. C. Preferred to forty-seven, and then the agents of Venus Equilateral stepped forth and offered to buy, at the market, all offered stock.

They did.

Then, as no more stock was offered, Interplanetary Communications Preferred rose sharply to ninety-four and stabilized at that figure. Terran Electric stock went through a valley, made by Kingman's sales, and then headed up, made by purchases on Terra, on Mars, and on Venus.

Don said: "Look, fellows, this has gone far enough. We have control again, and a goodly hunk of Terran Electric as well. Enough, I think, to force them to behave like a good little company and stay out of other people's hair. Let's all get together and celebrate."

"Right," echoed the men.

A month later, Joe's was the scene of a big banquet. Barney Carroll stood up and said:

"Ladies and gentlemen, we all know why we're here and what we're celebrating, so I won't have to recount the whole affair. We all think Don Channing is a great guy, and Walt Franks is not far behind, if any. I'm pretty likable myself, and my lifelong sparring partner Jim Baler is no smelt, either. And so on, ad infinitum.

"But, ladies and gentlemen, Don Channing has a dark, deep, dire, desperate phase of his life, one that he will be remembered and cursed for; one that

will weigh about his neck like a milestone—or is it millstone?—for all his life.

"Benefactor though he is, this much you shall know; I still say that there is no place in the inner system for a man who has made this possible. Listen!"

Barney raised his hand, and an attendant turned a standard, living room model radio receiver on. It burst into sound immediately.

"Ladies and gentlemen, the Interplanetary Network now brings to you the Whitewood Nutsies Program. Karven and Norwal, the Venusian Songbirds; Thalla; and Lillas, in person, coming to you from the jungles of Palanortis, on Venus, by courtesy of the Inter-planet Foods Co. of Battle Creek, Michigan!

"Ladies and gentlemen, Whitewood Nutsies are *GOOD* for you—"

Walt Franks said to Christine: "Let's get out of here."

Christine inspected Walt carefully. Then nodded. "Yup," she grinned. "Even you sound better than the Interplanetary Network!"

For once, Walt did not argue, having gained his point.

THE END

*** END OF THE PROJECT GUTENBERG EBOOK BEAM PIRATE ***

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